

Assignmmet 3

MPI / OpenMP Viva Questions and Answers

1. What is MPI?

MPI stands for Message Passing Interface.

It is a library used for parallel programming and communication between processes.

2. What is parallel computing?

Parallel computing means executing multiple tasks simultaneously using multiple processors or cores.

3. What is the aim of this practical?

To find the sum of N elements in an array using multiple processors.

4. What is OpenMPI?

OpenMPI is an open-source implementation of MPI used for distributed computing.

5. Why is MPI used?

MPI is used to reduce execution time by dividing work among multiple processors.

6. What is a process in MPI?

A process is an independent execution unit participating in parallel computation.

7. What is rank in MPI?

Rank is the unique ID assigned to each process.

Example:

```
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
```

8. What is MPI_COMM_WORLD?

It is the default communicator containing all MPI processes.

9. What is the purpose of MPI_Init()?

It initializes the MPI execution environment.

10. What is the purpose of MPI_Finalize()?

It terminates the MPI environment.

11. What is the purpose of MPI_Comm_size()?

It returns the total number of processes.

12. What is message passing?

Message passing is communication between processes using send and receive operations.

13. What is MPI_Send()?

MPI_Send() sends data from one process to another.

Example:

```
MPI_Send(&num, 1, MPI_INT, 1, 1, MPI_COMM_WORLD);
```

14. What is MPI_Recv()?

MPI_Recv() receives data from another process.

15. What is the role of rank 0?

Rank 0 usually acts as the master process.

16. What is a local sum?

Local sum is the partial sum calculated by each processor.

17. What is the final sum?

The final sum is obtained by adding all local sums together.

18. Why are arrays divided among processors?

To perform computations faster in parallel.

19. How many cores are used in the assignment program?

4 cores are used.

20. What is the total sum in the program output?

210

21. Which command is used to compile MPI programs?

mpicc filename.c

22. Which command is used to execute MPI programs?

mpirun -np 4 ./a.out

23. What does -np mean in MPI?

-np specifies the number of processes.

24. What language is commonly used with MPI?

C, C++, and Fortran.

25. What is distributed processing?

Distributed processing means dividing tasks among multiple processors or systems.

26. What happens if one process waits for a message?

It enters waiting/blocking mode until data is received.

27. What is the advantage of parallel execution?

It reduces time complexity and increases performance.

28. What is the difference between sequential and parallel execution?

Sequential

Parallel

One task at a time Multiple tasks simultaneously

Sequential

Parallel

Slower

Faster

Uses one processor Uses multiple processors

29. What is load distribution?

Dividing work equally among processors.

30. What is the outcome of this practical?

Successfully implemented distributed sum calculation using MPI.